

Data Vortex - Aaruush '26

Round 1 Phase 2 - SQL Query Submission

Team: Punch Queries **Members:** Sahil Patel, Suhani Gupta

Selected questions: E1 (Easy), M5 (Medium), H3 (Hard)

Submission note: run these queries in SQL Server / SSMS against the DataVortex database. The repository's final schema defines `dbo.posts.engagement` as a persisted computed column. The final `dbo.posts` table does not contain `shares_exceed_likes`, so M5 evaluates its condition directly.

E1 - Platform Popularity

Question

Determine which social media platform has the highest number of posts. Ignore posts where the platform is missing. Return the platform name and number of posts.

Final SQL

```
USE DataVortex;
GO

/* E1 - Platform Popularity
   Unknown represents missing platform values in the cleaned dataset.
   Exclude it before counting. */
SELECT TOP (1)
    platform,
    COUNT(*) AS post_count
FROM dbo.posts
WHERE platform <> 'Unknown'
GROUP BY platform
ORDER BY COUNT(*) DESC, platform ASC;
GO
```

M5 - Detect Suspicious Engagement

Question

Find posts where the number of shares is greater than the number of likes and comments combined. Return the top 20.

Final SQL

```
USE DataVortex;
GO

/* M5 - Detect Suspicious Engagement
   The final dbo.posts table does not store shares_exceed_likes as a
   persisted column; that flag exists in the cleaned/staging data.
   Evaluate the questionnaire condition directly from base metrics. */
SELECT TOP (20)
    post_id,
    platform,
    likes,
    shares,
    comments,
    engagement
FROM dbo.posts
WHERE shares > likes + comments
ORDER BY engagement DESC, post_id ASC;
GO
```

H3 - Platform Performance Compared With Its Own Average

Question

For every platform, identify posts whose engagement is significantly higher than the average engagement of that platform. A post is exceptional if its engagement is at least 2x the average engagement of its platform.

Final SQL

```
USE DataVortex;
GO

/* H3 - Platform Performance Compared With Its Own Average
   Compute each platform's average engagement with a window function,
   then keep posts at or above 2x that platform-specific average. */
WITH scored_posts AS
(
    SELECT
        p.post_id,
        p.user_id,
        p.platform,
        p.likes,
        p.shares,
        p.comments,
        p.engagement,
        AVG(CAST(p.engagement AS FLOAT))
            OVER (PARTITION BY p.platform) AS platform_avg_engagement
    FROM dbo.posts AS p
)
SELECT
    post_id,
    user_id,
    platform,
    likes,
    shares,
    comments,
    engagement,
    CAST(platform_avg_engagement AS DECIMAL(10,2))
        AS platform_avg_engagement,
    CAST(2 * platform_avg_engagement AS DECIMAL(10,2))
        AS exceptional_threshold
FROM scored_posts
WHERE engagement >= 2 * platform_avg_engagement
ORDER BY platform ASC, engagement DESC, post_id ASC;
GO
```